Robust FLOWER Under Noise Model Mismatch

A controlled study of reconstruction quality

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Generative Learning and Diffusion Models

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FLOWER: From a Measurement to a Clean Image

Inverse Problem

We observe a degraded, noisy image and seek the underlying clean image.

The Image Prior

A pretrained flow model predicts a clean destination from the current trajectory state.

$$y = Hx + n, \qquad n \sim \mathcal{N}(0, \sigma_{\text{true}}^2 I)$$

Three Steps at Every Flow Time

1. Estimate the clean destination
2. Refine it using the measurement
3. Move to the next flow time

The Noise Level Sets the Refinement Strength

Destination Estimate

$$X_t = (1-t)X_0 + tX_1, \qquad \hat{x}_1(x_t) = x_t + (1-t)v_\theta(x_t, t)$$

Measurement-Aware Refinement

$$\mu_t = \arg\min_x \left\{ \frac{\|Hx-y\|_2^2}{2\sigma_a^2} + \frac{\|x-\hat{x}_1\|_2^2}{2\nu_t^2} \right\}$$

Interpretation

A smaller assumed σ gives more weight to the measurement.
A larger assumed σ gives more weight to the predicted destination.

The paper's image benchmark uses Flower1-OT with γ = 0

Our Research Question

Noise Calibration

How does reconstruction quality change when FLOWER receives an incorrect noise standard deviation?

$$r = \sigma_{assumed} / \sigma_{true}$$

r = 1 is the correct-noise control

r	0.50	0.75	1.00	1.25	1.50
Variance ratio	0.25	0.5625	1.00	1.5625	2.25

Paired Comparison

Same image, measurement and source-noise draws for every ratio. True σ and noise structure stay fixed. Only assumed σ changes.

Experimental Design

20 fixed CelebA images + 20 fixed AFHQ-Cat images

Only assumed σ changes. Fixed image, H, measurement and noise structure. γ = 0.

Inverse problem	True σ	CelebA	AFHQ-Cat
Denoising	0.20	Identity	Identity
Gaussian deblurring	0.05	61 × 61, blur σ = 1	61 × 61, blur σ = 3
Super-resolution	0.05	2× subsampling	4× subsampling
Random inpainting	0.01	70% missing	70% missing
Box inpainting	0.05	40 × 40 central box	80 × 80 central box

40 images × 5 problems × 5 ratios = 1,000 reconstructions

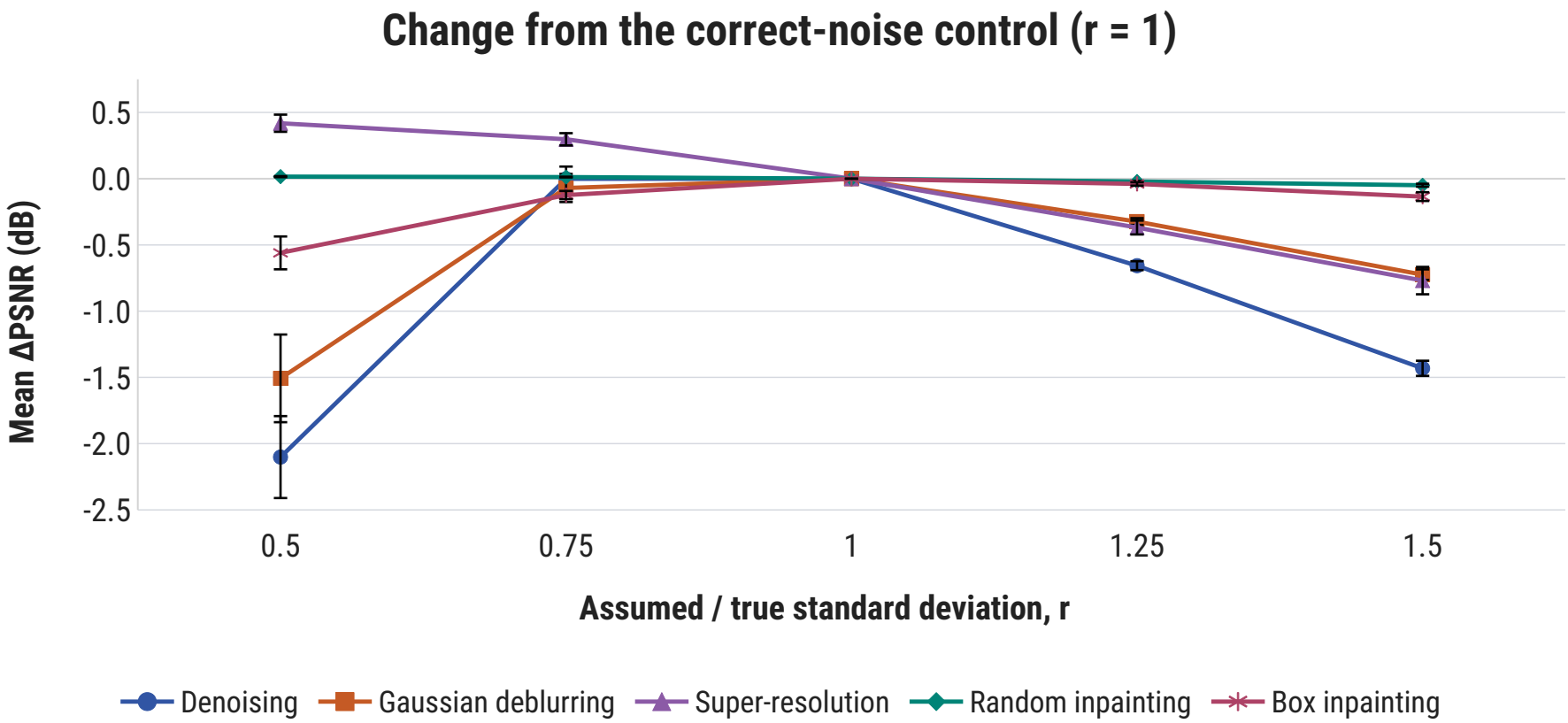
PSNR ↑ SSIM ↑ LPIPS ↓

$$\Delta M(r) = M(r) - M(1)$$

Zero = no change. PSNR / SSIM: positive is better. LPIPS: negative is better.

Same selected images throughout each dataset. Seed 2026. Noise σ is in model coordinates [-1, 1].

PSNR Response: CelebA

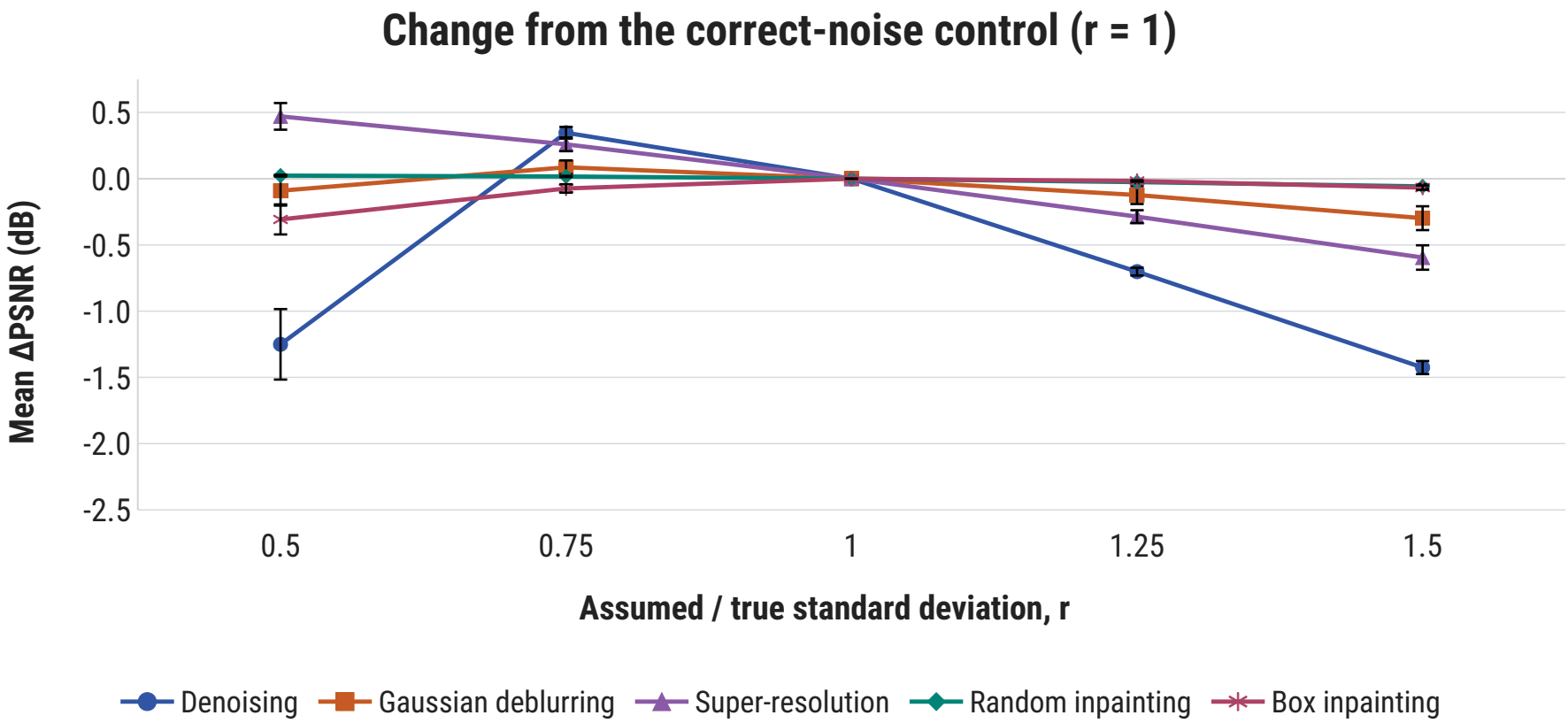


Main Observation

Denoising loses 2.10 dB at r = 0.50. Super-resolution gains 0.42 dB. Random inpainting changes very little.

ΔPSNR = PSNR(r) − PSNR(1). Zero = no change. Higher is better. n = 20. Error bars: approximate 95% CI.

PSNR Response: AFHQ-Cat

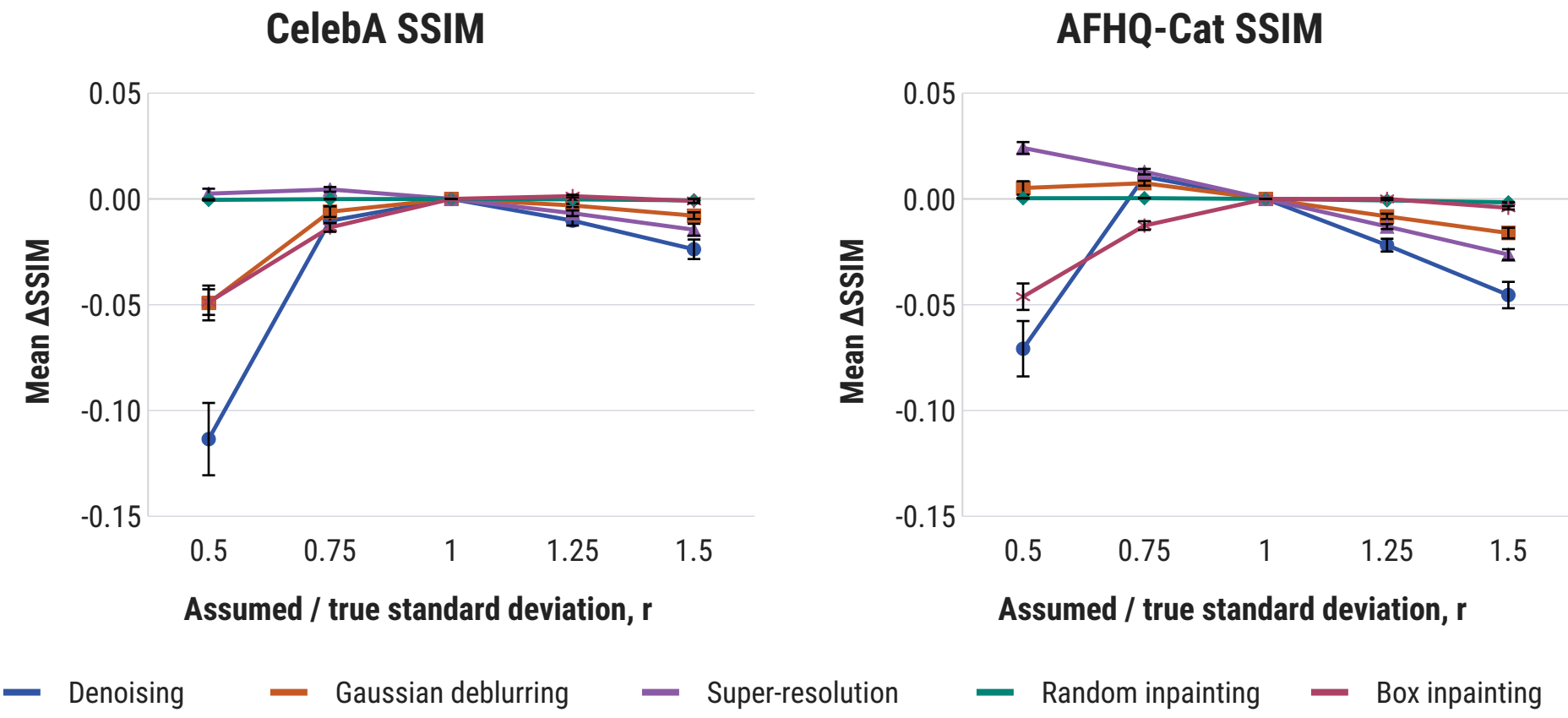


Main Observation

Denoising improves by 0.35 dB at r = 0.75, but loses 1.25 dB at r = 0.50. Super-resolution gains 0.47 dB at r = 0.50.

ΔPSNR = PSNR(r) − PSNR(1). Zero = no change. Higher is better. n = 20. Error bars: approximate 95% CI.

SSIM Response Across Both Datasets

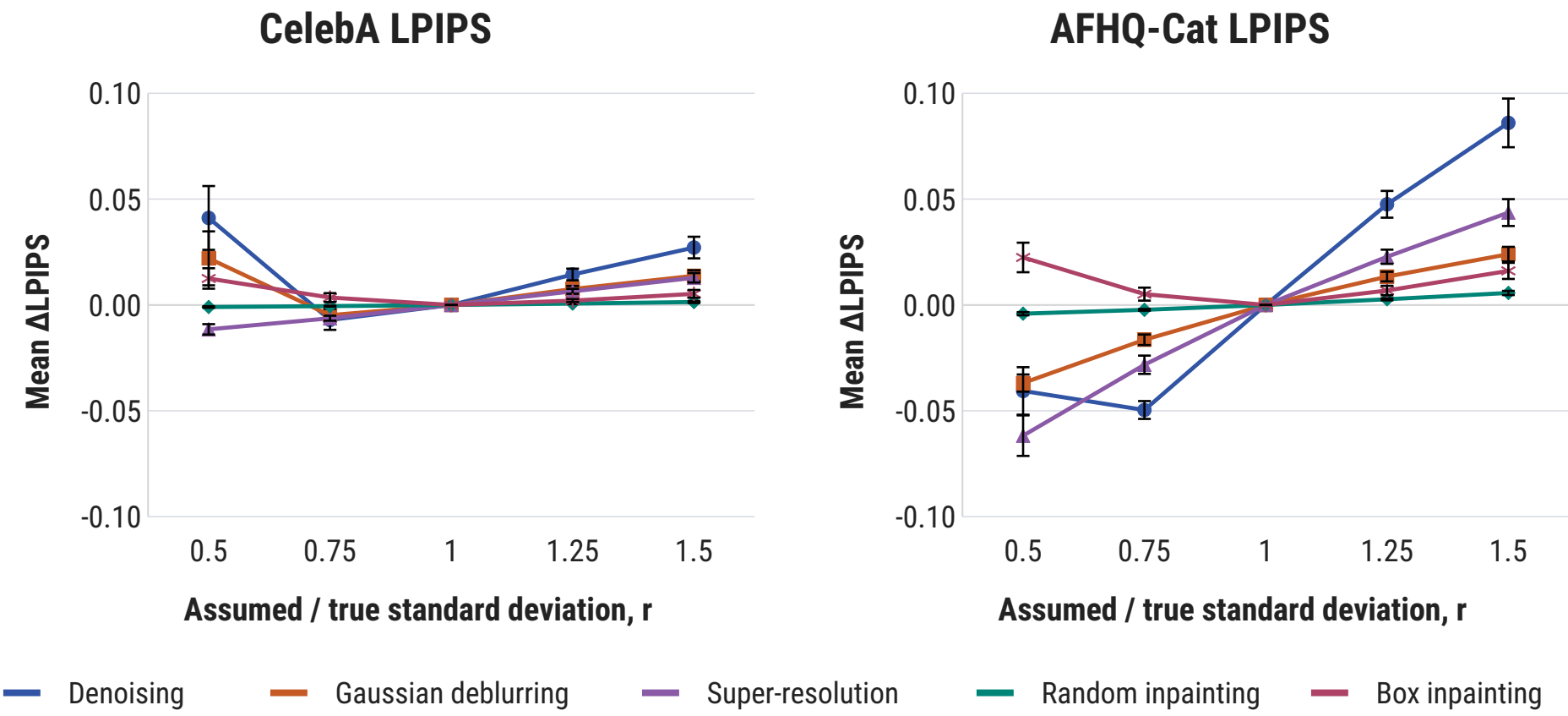


What the Metric Adds

Underestimating noise can damage structure even when PSNR changes little. Super-resolution improves in both datasets.

Mean paired change vs r = 1. Higher is better. n = 20. Error bars: approximate 95% CI.

LPIPS Response Across Both Datasets



What the Metric Adds

At r = 0.50, AFHQ-Cat denoising improves LPIPS while PSNR and SSIM worsen. Perceptual and pixel fidelity can disagree.

Mean paired change vs r = 1. Lower is better. n = 20. Error bars: approximate 95% CI.

All Tasks: Mean PSNR Change (dB)

Cells compare a mismatched reconstruction with its r = 1 control

CelebA

Task	0.50	0.75	1.25	1.50
Denoising	-2.10	0.00	-0.66	-1.43
deblurring	-1.51	-0.07	-0.32	-0.72
Super-resolution	+0.42	+0.30	-0.37	-0.77
Random inp.	+0.02	+0.01	-0.02	-0.05
Box inp.	-0.56	-0.12	-0.04	-0.14

AFHQ-Cat

Task	0.50	0.75	1.25	1.50
Denoising	-1.25	+0.35	-0.70	-1.43
deblurring	-0.09	+0.09	-0.12	-0.30
Super-resolution	+0.47	+0.26	-0.29	-0.60
Random inp.	+0.02	+0.02	-0.03	-0.06
Box inp.	-0.31	-0.07	-0.02	-0.07

Reading the Table

Positive values mean better PSNR. Overestimation lowers the mean in every tested dataset–task setting.

Each cell averages 20 paired image differences. The r = 1 column is zero by definition.

Visual Comparison: Denoising

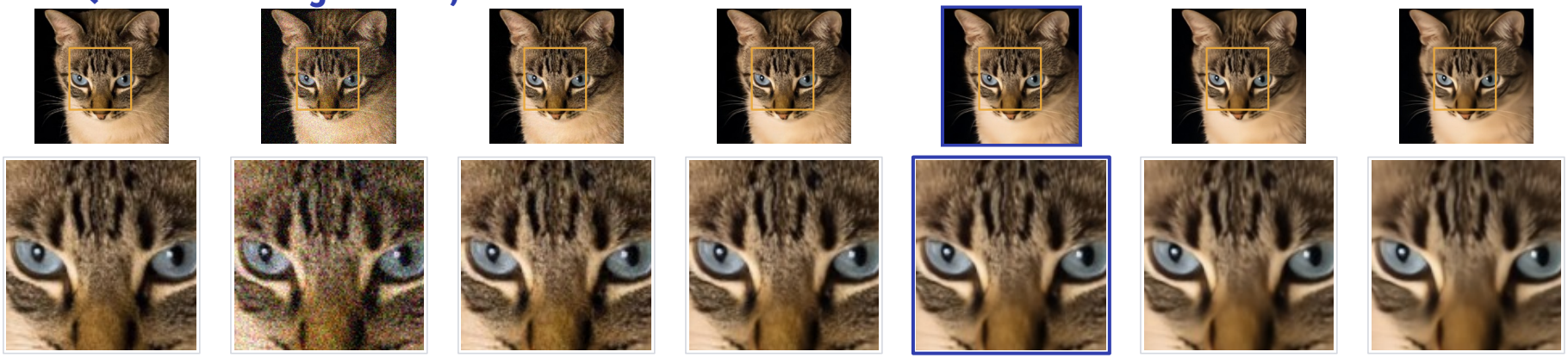
Ground truth	Measurement	r = 0.50	r = 0.75	**r = 1.00**	r = 1.25	r = 1.50

CelebA: full image above, 3× detail below



		29.72 / 0.803 0.097	31.77 / 0.913 0.033	31.85 / 0.928 0.020	31.34 / 0.920 0.028	30.70 / 0.909 0.037

AFHQ-Cat: full image above, 3× detail below

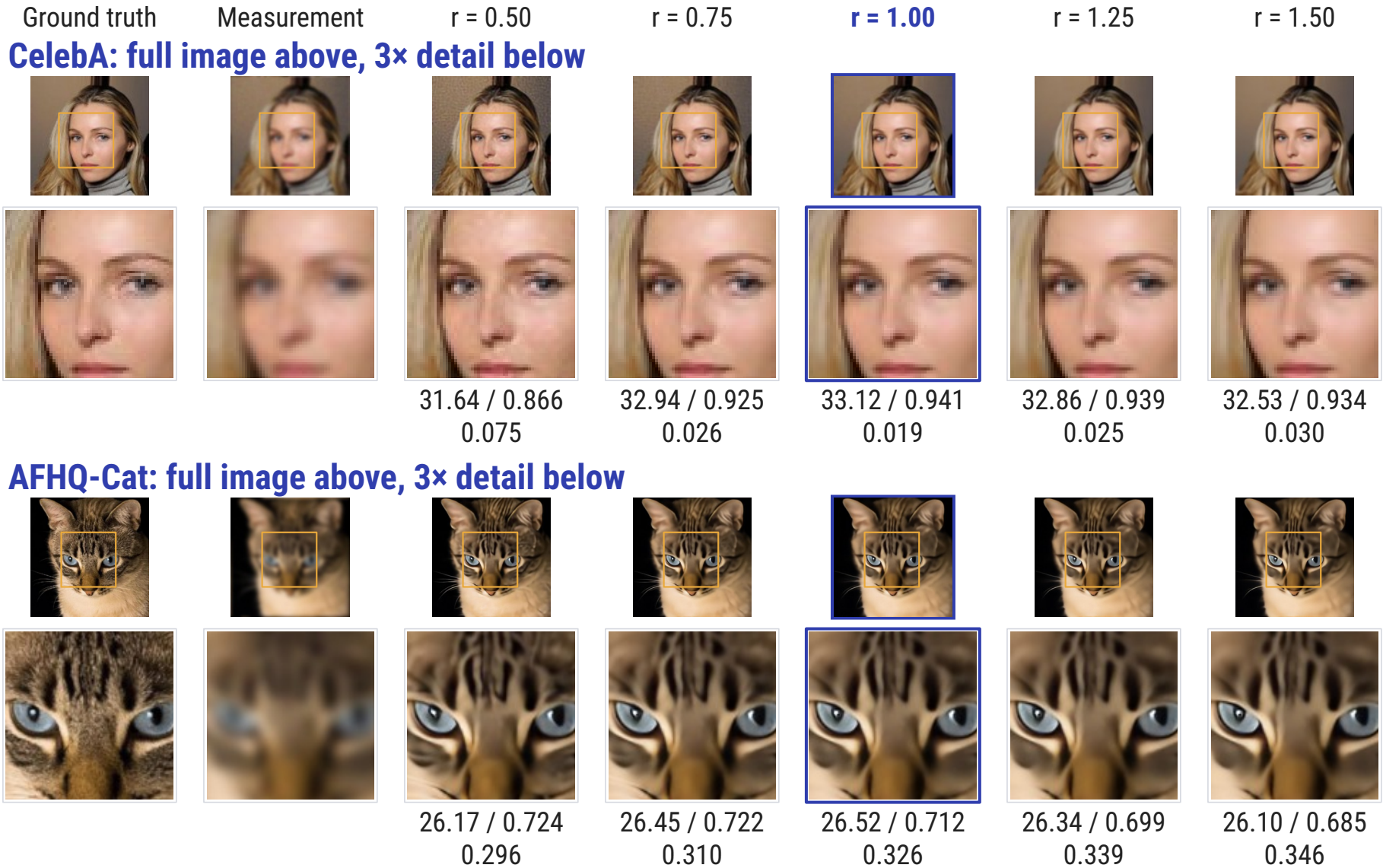


		29.78 / 0.813 0.043	30.87 / 0.868 0.057	30.57 / 0.846 0.111	29.89 / 0.812 0.162	29.15 / 0.774 0.203

Finding Lower assumed noise leaves more fine-scale variation. Higher assumed noise smooths it.

Metrics: full-image PSNR (dB) / SSIM, then LPIPS. Gold box: crop region. Blue frame: r = 1. Measurement: $H^{T}y$.

Visual Comparison: Gaussian deblurring



Finding The two domains respond differently. Blur kernels and image content also differ.

Metrics: full-image PSNR (dB) / SSIM, then LPIPS. Gold box: crop region. Blue frame: r = 1. Measurement: $H^{T}y$.

Visual Comparison: Super-resolution

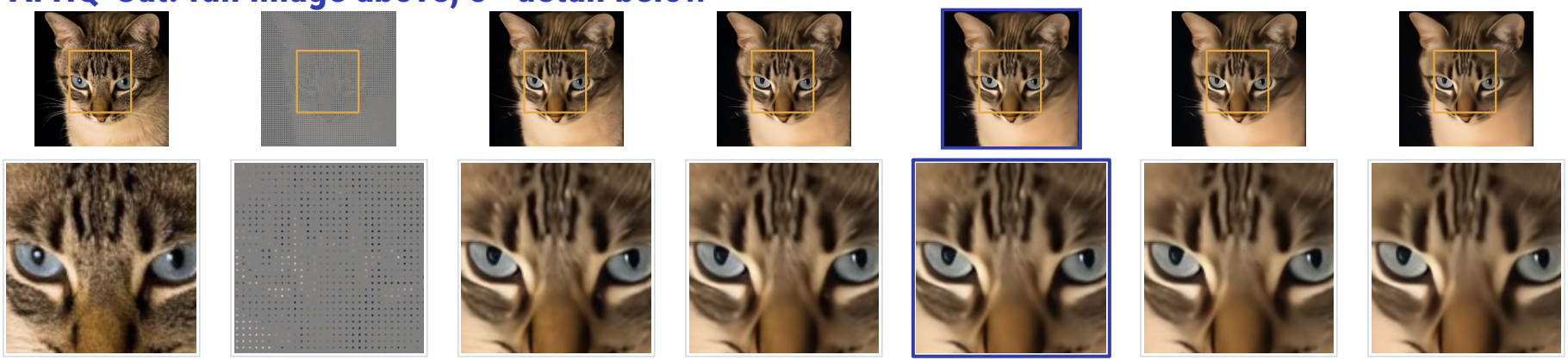
Ground truth	Measurement	r = 0.50	r = 0.75	**r = 1.00**	r = 1.25	r = 1.50

CelebA: full image above, 3× detail below



		31.34 / 0.921 0.016	31.29 / 0.924 0.018	31.13 / 0.922 0.022	30.90 / 0.917 0.026	30.64 / 0.911 0.030

AFHQ-Cat: full image above, 3× detail below

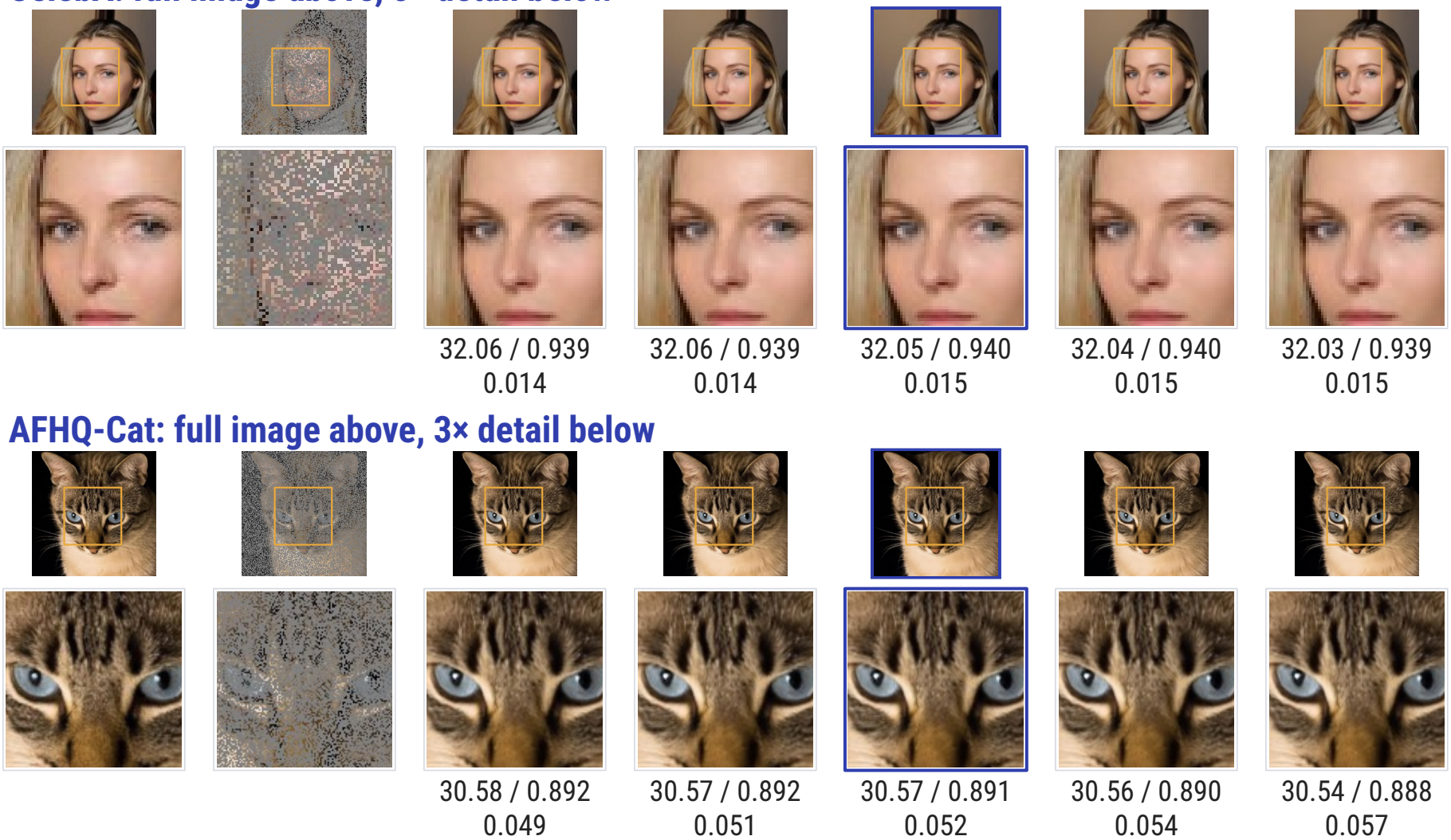


		25.57 / 0.663 0.247	25.46 / 0.643 0.271	25.33 / 0.624 0.292	25.16 / 0.606 0.308	24.97 / 0.590 0.324

Finding At r = 0.50, both datasets improve in all three metrics on average across 20 images.

Metrics: full-image PSNR (dB) / SSIM, then LPIPS. Gold box: crop region. Blue frame: r = 1. Measurement: $H^{T}y$.

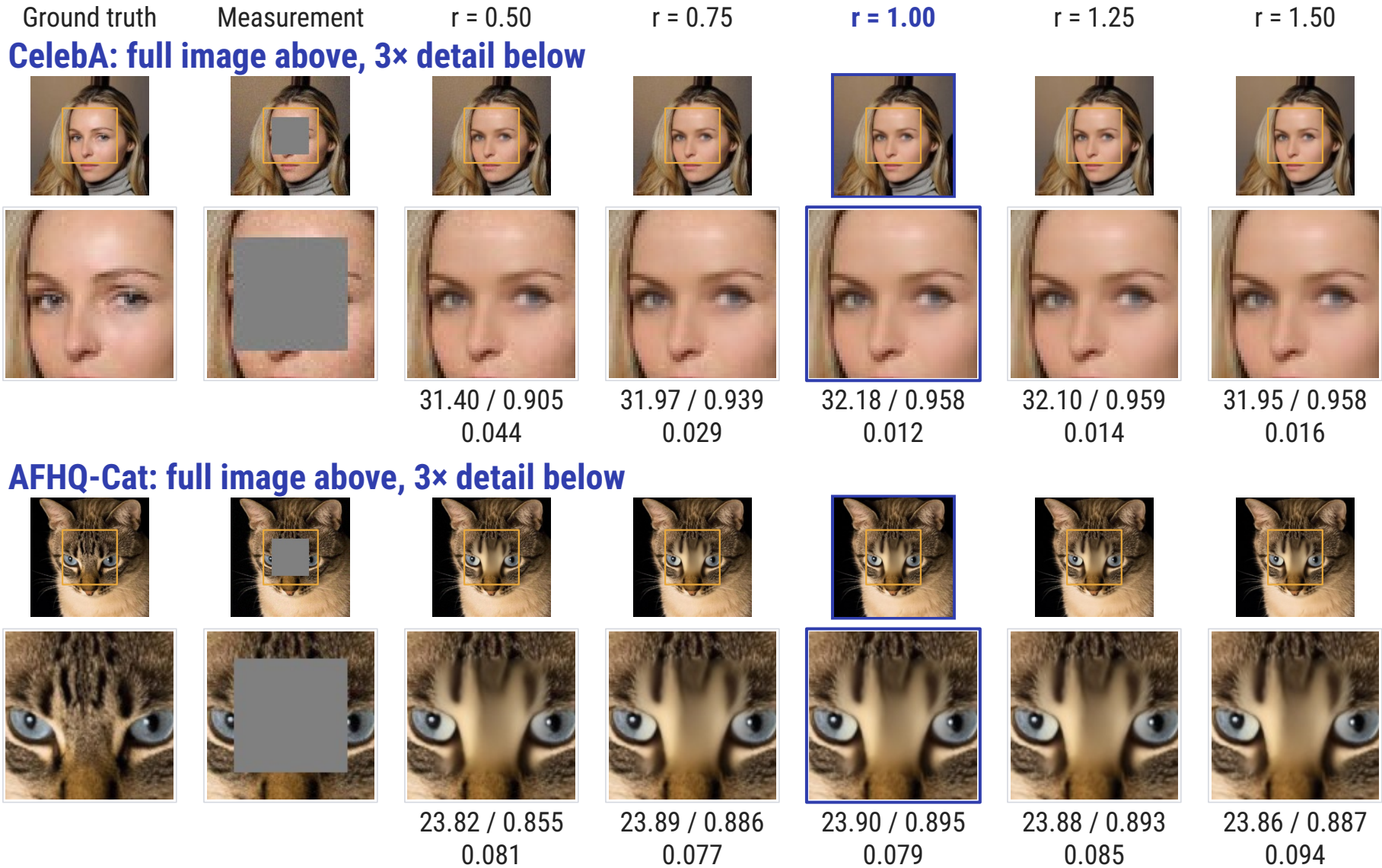
Visual Comparison: Random inpainting



Finding Changes are small across this range. This task also has the lowest true σ, 0.01.

Metrics: full-image PSNR (dB) / SSIM, then LPIPS. Gold box: crop region. Blue frame: r = 1. Measurement: $H^{T}y$.

Visual Comparison: Box inpainting

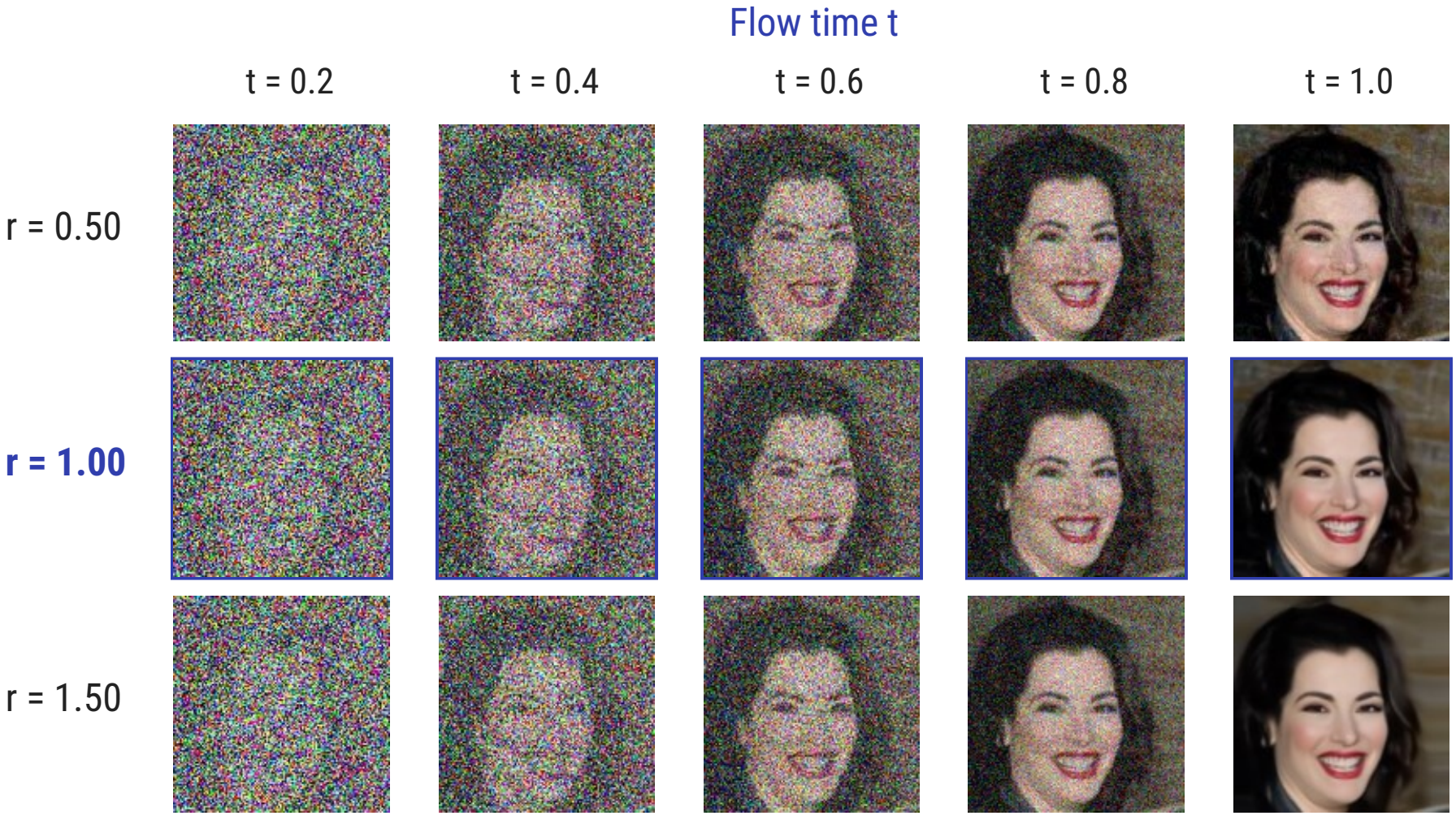


Finding The central region is unobserved. The prior and observed pixels jointly guide its reconstruction.

Metrics: full-image PSNR (dB) / SSIM, then LPIPS. Gold box: crop region. Blue frame: r = 1. Measurement: $H^{T}y$.

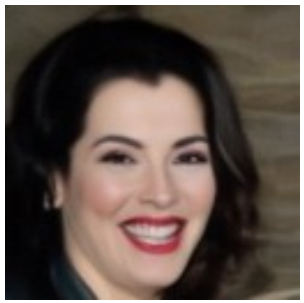
How Mismatch Changes the Trajectory

CelebA denoising, image 182734.jpg



Rows share measurement and source-noise draws. Final column: reconstruction. All five ratios are in the report.

Inside the Three Steps at t = 0.75

	Destination estimate	Measurement refinement	Next trajectory state
r = 0.50			
r = 1.00			
r = 1.50			

CelebA denoising, image 182734.jpg. Step 2 uses σ directly. Step 3 adds source noise and advances time.

Interpreting the Evidence

Possible Explanation: Measurement Constraints

Denoising constrains every pixel. Masking and subsampling constrain observed locations directly. True noise levels also differ across tasks.

Possible Explanation: Approximation Error

A different measurement weight may partly compensate for prior or solver approximation errors. The experiment does not establish why underestimation sometimes improves the metrics.

Limits of the Comparison

20 images per dataset versus 100 in the paper. One paired seed sequence per image and task. Task and true noise level vary together.

These explanations are hypotheses. The measured effects apply to the tested sample, tasks and noise ratios.

Conclusions

FLOWER shows task-dependent robustness to an incorrect assumed noise level

Sensitivity Depends on the Task

Denoising has the largest PSNR losses. Random inpainting changes very little in the tested settings.

Overestimation Consistently Hurts Mean PSNR

Both r = 1.25 and r = 1.50 reduce mean PSNR in all ten tested dataset–task settings.

Underestimation Has No Universal Outcome

Underestimation improves super-resolution here, but strong underestimation hurts denoising. LPIPS can disagree with PSNR and SSIM.

Conclusions apply to the tested images, settings and noise ratios.

References and Reproducible Results

M. Pourya, B. El Rawas and M. Unser
FLOWER: A Flow-Matching Solver for Inverse Problems
ICLR 2026

Y. Lipman et al.
Flow Matching for Generative Modeling
ICLR 2023

Project Results

Raw and paired metrics, all 50 summary rows, saved examples, and trajectory captures are in artifacts/experiments/main/.

github.com/ItayShorian07/Diffusion--flower-project